YUANJUN (JOHNNY) DAI

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SUMMARY

AI Infrastructure and Applied AI Engineer, **PhD candidate** at CWRU. Specializes in the security, optimization, and observability of ML/LLM training and inference infrastructure. 8 research papers (7 first-author) at IEEE, ACM, and Springer; **2+ years of systems software engineering** at Cisco and Broadcom; 2+ years of **cross-functional** product ownership at Midea and Xiaomi.

EDUCATION

Ph.D. Candidate in Computer Science (AI Infrastructure) — 08/2019 – Present
Case Western Reserve University

M.S. in Electrical and Computer Engineering (Computer Architecture)
University of Pittsburgh

B.S. in Electrical Engineering
Northeastern University

SKILLS

Languages	Python, C/C++
ML Frameworks	PyTorch (DDP/FSDP), vLLM, DeepSpeed, Megatron-LM, NCCL, BytePS, scikit-learn
Parallelism Paradigms	Data Parallel (DDP/FSDP), Model Parallel (TP/PP), Pipeline Parallel (DeepSpeed), Hybrid Parallel
Infrastructure	Kubernetes, Docker, CI/CD Pipelines, ArgoCD, Slurm, HPC
Systems & APIs	FastAPI, REST API, gRPC
Observability	eBPF (Kernel Tracing), Prometheus, Grafana, DCGM Exporter, Intel RAPL
Cloud & HPC	Lambda Labs HPC, Ohio Supercomputer Center, NSF FABRIC, CloudLab
Networking	SDN (Software-Defined Networking), P4, NPU/ASIC (Broadcom BCM SDK)

RESEARCH & SYSTEMS ENGINEERING EXPERIENCE

Research Assistant, AI Infrastructure & ML Systems — 08/2019 – Sep. 2026 (exp.)

Case Western Reserve University

Research focus: optimization, observability, and security of distributed ML/LLM training and inference infrastructure.

- **DYNAMIX – RL-Based Adaptive Batch Scheduling for Distributed ML** | Paper / Code

  Designed a **PPO**-based RL scheduler monitoring hardware signals (CPU/memory, bandwidth, retransmission rate via custom **eBPF** probes) and statistical signals (loss convergence, generalization) to dynamically adapt batch size; co-scales learning rate via **Linear Scaling Rule**; evaluated on **Ring AllReduce (DDP)** and **Parameter Server (BytePS)** across **8–64 A100 GPUs**; achieves **46% end-to-end training time reduction** and improved final model accuracy over fixed batch size baseline.

- **PRISM – Priority-Based Selective Reliability for Distributed ML Training** | Paper (Under Review)

  Designed a transport-layer library (**NCCL/Gloo → UDP**) hooking into **PyTorch DDP**'s AllReduce at bucket-ready events; classifies gradients by magnitude into high-priority (full **SACK**-based retransmission) and low-priority (local compensation via cached values); supports kernel-bypass (**LibVMA**) and kernel-based (**GSO/GRO**) variants; achieves **8–15% throughput gain for CNNs and 15–21% for LLMs** over TCP, matching NDP latency at 144-worker scale.

- **Scalable LLM Serving Infrastructure on Kubernetes** | Code

  Independently designed and built end-to-end a **K8s** microservices platform (Gateway / Router / Worker) with **vLLM** inference, **ArgoCD** GitOps, **Prometheus/Grafana** observability, and **CI/CD** across **50+ deployments**.

- **Domain-Adapted LLMs for Clinical Risk Adjustment Coding** | Paper

  Designed chronic disease classifier (115 HCC codes) on extremely imbalanced multi-label EHR data (MIMIC-V); built semi-supervised labeling via **few-shot GPT-4o** prompting; applied **Fast Attention** ($O(Ld)$ vs $O(L^2)$), inverse-frequency per-label weighting, and per-label threshold tuning; fine-tuned **PubMedBERT** (**NVIDIA NeMo**) and **Qwen-7B** (**Megatron-LM + DeepSpeed**) on Lambda Labs A100; achieves **macro-/micro-F1 of 0.775/0.742** and macro-AUC of 0.9362.

- **Priority-Based eBPF Network Flow Telemetry for AI / DML Infrastructure** | Paper / Code

  Designed PSketch, the first in-kernel priority-aware sketch using **eBPF**; high-priority flows tracked via BPF_HASH ($O(1)$), low-priority via **3-layer Count-Min Sketch**; voting-based collision resolution and atomic operations for thread-safe in-kernel updates; achieves **96% Top-K accuracy, 96.4% retransmission recall**, and <1.1% throughput overhead at 10 Gbps.

- **High-Resolution eBPF System Resource Profiling for DML Workloads** | Paper / Code

  Designed eHashPipe, a dual-pipeline **eBPF** framework: (1) HashPipe-based memory profiler with cascading slot eviction and ptr2size deallocation tracking, (2) per-thread CPU tracker via sched_switch tracepoint; delivers **10–14× finer temporal resolution** than top/htop, ~11ms latency, and >90% Top-K fidelity.

- **Training-Time Side-Channel Attacks on Distributed ML Systems** | Paper / Code

  Exposed MLaaS training-phase side-channel vulnerability via deep analysis of DML computation/communication behavior in Parameter Server architecture; applied **K-means clustering** on push/pull traffic (silent period detection), **Fourier transform** on **Intel RAPL** for noise reduction, **Decision Tree** for layer-type classification, and **polynomial regression** for **activation function profiling**; achieves **>99% layer-type accuracy and <1.5% tensor-size error** across ZFNet, AlexNet, and VGG.

- **Scotch – Elastic SDN Overlay for Control-Plane Scalability under DDoS** | Paper / Code

  Designed an **OvS**-based elastic overlay: when physical switch OFA is saturated, tunnels new flows to a **vSwitch** pool via **event-driven** Packet-In Edge messages; separates small DDoS flows (terminated at vSwitches) from large flows (migrated to physical paths); achieves **>70% controller congestion reduction** on 34-node GENI Clos topology without hardware modification.

INDUSTRY EXPERIENCE

Software Engineer II
06/2015 – 08/2016

Cisco Systems, Inc.

- Developed NPU/ASIC features for a 100 Gbps core router (Azure, AWS, Alibaba Cloud) using Broadcom BCM SDK.
- Contributed to QoS and ECMP modules, reducing forwarding latency under heavy load.

Software Engineer Intern
10/2014 – 05/2015

Broadcom (Emulex)

- Replaced polling-based data acquisition with DMA + circular-buffer architecture in an RTOS pipeline (**30% CPU load reduction**); ported BladeEngine ASIC management CLI from Linux to Windows Server.

Product Manager
09/2016 – 06/2018

Midea Group

- Led end-to-end technical PM for **Fun Oven II** — overseeing ML model training & deployment, Docker services on Alibaba Cloud, mobile app integration, and culinary algorithm design; product won the **Red Dot Design Award** and debuted at 2018 AWE Show · Red Dot ↗ · PR Newswire ↗

Technical Product Manager
07/2018 – 01/2019

Xiaomi

- Defined product requirements for a smart IoT appliance via competitive analysis and user research; assessed technical feasibility and delivered architecture proposals to align engineering and business stakeholders.

Engineer Intern
01/2014 – 04/2014

Tenova I2S

- Re-architected an RTOS data acquisition pipeline using DMA and circular buffers, **reducing CPU load by 42%** and improving sampling rate and control-loop responsiveness.

PUBLICATIONS

8 papers total, 7 first-author, 5 published at IEEE CCNC, IEEE ICNC, ACM TOIT, and Springer SecureComm; 3 under review (incl. ACM TOPS). Full list: johnny-dai-git.github.io/portfolio